

Radiation Detector Integration with Real-Time Data Recorder Application

Ethan Hartman
[IMAGE OF PROTOTYPE]

Technical Field

This project involves the technical fields of Embedded systems with integrated sensors, Application development, Data Management, Statistical Data Science, Kotlin, C/C++.

Background Information

Drawing from nearly a decade of operational experience and challenges in a naval nuclear environment, the project was conceived to address the critical shortcomings in current radiation survey methodologies. While modern monitoring equipment is effective and provides some mobility, it lacks ease of data logging and intuitive technological integration desired by field technicians, particularly in the US Naval Nuclear Propulsion Program. Inspired by these needs, this initiative combines embedded sensor technology with an offline-enabled tablet application to offer real-time, map-based data visualization and archival. This innovative approach aims to combine and enhance safety, accuracy, and efficiency of radiation surveys in environments where conventional, bulky devices are inadequate and where storage space for archived paper logs is limited.

Prior Art (Research)

Prior Art Disclaimer

There is prior art in the field of portable radiation detection devices. Although these existing solutions serve as valuable references for measuring radiation levels, they do not provide the cohesive offline mapping, integrated data logging, and archiving functionality proposed in this project. The following product illustrates how current technologies address core detection needs but leave gaps that are fulfilled by this innovation.

The RadEye PRD

A compact, highly sensitive personal radiation detector designed for quick deployment in various radiation monitoring scenarios. It offers audible and visual alarms and can store basic reading data.

Reference:

Thermo Fisher Scientific. (n.d.). *RadEye PRD personal radiation detector*.
<https://www.thermofisher.com/order/catalog/product/425069172>

Project Description

This project focuses on creating a handheld radiation detection system for nuclear propulsion technicians and private-sector radiological technicians, typically 25–50 years old and tech-savvy, who often work in confined spaces requiring accurate, real-time measurements. A Geiger-Müller tube and Matrix Portal S3 microcontroller (programmed in C/C++) capture and log radiation data, which is transferred via a wired connection to a Kotlin-based Android tablet application. The tablet provides a secure, offline mapping interface where users can select locations, store sensor readings in a local SQLite database, and view real-time markers for immediate feedback. By streamlining data recording and auditing, the system aims to reduce administrative overhead and enhance operational efficiency in radiation surveys.

Innovation Claim

This solution is an improvement to existing designs because it provides a highly mobile, easy-to-use, and cost-effective alternative to bulky or proprietary systems. It enhances data accessibility for technicians conducting manual radiation surveys, ensuring accuracy and reliability.

Usage Scenario

In a naval nuclear power plant, the traditional reliance on paper logs for radiological surveys in confined spaces poses significant challenges, including limited physical storage, time-consuming data transfer, and the heightened risk of data loss or damage. The innovative handheld radiation detection system paired with an offline tablet application offers a transformative solution by digitally capturing real-time radiation readings and annotating them directly onto a secure facility map. This advanced approach eliminates the cumbersome manual process, dramatically reduces transcription errors, and ensures that safety-critical data is reliably archived in a local SQLite database. By streamlining administrative tasks and enhancing data integrity, the system not only improves operational efficiency but also significantly bolsters compliance with stringent safety protocols, ultimately providing a highly reliable tool for radiological maintenance in high-security environments.

SIP BRIEF PART 2

Evaluation Criteria

Does the Matrix Portal S3 successfully read the current generated by the Geiger-Muller tube and convert it to a measurable exposure rate?

Can the Android application generate a new map, and does it allow the user to save and load maps?

Is the current exposure rate reliably displayed on the Android application in real time?

Can the user log multiple types of surveys (radiation, airborne, contamination) onto the map?

Can the survey map be saved, restored with all survey data intact, and edited without loss of information?

Does the system operate entirely offline via a wired connection without the need for wireless communication?

Goals and Tasks Associated with the Project

Objective 1: Integrate Reliable Sensor Data Capture

Tasks:

- Develop C/C++ firmware on the Matrix Portal S3 to accurately read pulses from the Geiger-Müller tube.
- Implement an algorithm to convert pulse counts into a corresponding exposure rate.
- Establish and verify a stable, wired USB communication link to transmit data to the Android tablet.
- Validate sensor performance through controlled tests to confirm that the generated exposure rate is accurate and consistent.

Objective 2: Develop a Secure, Offline-Capable Android Application

Tasks:

- Design and implement a user-friendly interface that features an interactive map for selecting survey locations.
- Incorporate functionality for users to generate, save, load, and edit survey maps.
- Integrate a local SQLite database to store all sensor data and survey logs securely.

Description of Design Prototype

The design prototype for this project will consist of a custom-built radiation survey and mapping application paired with a portable radiation detection device. The system is intended to emulate the functionality and workflow of survey documentation systems used in industrial and nuclear environments while modernizing the process through digital mapping, data persistence, and automated survey organization.

The prototype will be developed primarily for Android tablets using Kotlin in Android Studio. The tablet application will function as the central user interface and data management system for the project. The radiation detector hardware will be built around a microcontroller-based platform capable of collecting and transmitting radiation count data to the tablet application over a wired USB connection. SQLite will be used as the local database solution to store map templates, survey records, metadata, and associated survey point information directly on the device without requiring internet connectivity. The application is intentionally designed as an offline-first system due to the secure and isolated environments where radiation surveys are commonly performed.

The prototype application will contain several major functional systems that collectively define the project. One major component is the template map management system. Users will be able to upload facility map images directly into the application and save them as reusable templates. These templates may then be filtered and selected based on categories such as area, room, floor, or survey type. This functionality is important because radiation survey work is highly location-dependent and often requires repeated use of the same facility layouts over time.

Another major component is the interactive survey mapping system. Once a map template is selected, the user will be able to place survey points directly onto the map using

touch interaction. Different survey types such as radiation, contamination swipes, airborne samples, and frisk points will use distinct symbols and numbering systems. Certain survey types will automatically increment identifiers sequentially to maintain procedural consistency. Survey data points will contain associated information such as dose rate, contamination activity, timestamps, comments, and surveyor metadata.

The prototype will also demonstrate persistent data storage and retrieval. Completed surveys can be saved and later reloaded with all associated map points and metadata intact. Users will also be able to edit previously saved maps and continue surveys without losing existing information. To improve performance, flattened front and back survey images may also be generated and stored for rapid viewing while preserving the underlying editable data.

The hardware portion of the prototype will demonstrate radiation data acquisition and communication. A Geiger-Müller tube-based detector connected to a microcontroller will generate count data which can either be simulated or transmitted live to the Android application. The application will display current readings and associate them with selected survey points. Early prototype stages may use simulated data streams before transitioning to live hardware integration.

To run the prototype, the user will install the Android application onto a compatible Android tablet or execute it through the Android Studio emulator. The application will launch into a home screen where users may upload maps, select templates, or open saved survey records. If hardware integration is enabled, the detector device will connect to the tablet through USB and stream radiation data into the application automatically. The prototype is intended to demonstrate the complete workflow of selecting a facility map, recording survey data onto the map, saving the survey, and later retrieving or editing that information.

This design prototype directly supports the final project goals by validating the system architecture, user workflow, storage strategy, and hardware-software integration approach. The prototype focuses on proving the practicality and usability of the system while establishing the technical foundation for a more polished and production-ready implementation in the completed project.

Project Timeline Completion

Completed:

2026-05-27: Established the Compose UI shell, Room database scaffolding, repository wiring, and baseline map workflow.

2026-05-28: Completed survey mode selection, front/back page handling, constrained point placement, erase support, and draft-first state management.

2026-05-30: Implemented front/back rendering, editable contamination and airborne table workflows, and expanded survey metadata.

2026-06-02: Validated robust save/load behavior, draft commits, bulk persistence for survey data, blank-map rendering, and metadata propagation.

2026-07-01: Added a canvas-based map creator with shape tools and PNG template output.

2026-07-05: Separated reusable map templates from survey records and allowed records to reopen specific saved surveys.

2026-07-10: Enforced template-only draft behavior until Save Job, moved notes/signatures to survey scope, and updated Room schema to version 5.

2026-07-12: Expanded canvas tools with square, ellipse, triangle, diamond, line, arrow, and door/opening options.

2026-07-16: Made the canvas shape palette fully visible and added scrolling so map creation tools remain reachable on smaller screens.

2026-07-18: Aligned Home, Stored Maps, and Records with the planned multi-panel layouts and added record preview image support.

2026-07-19: Added shared visible scroll rails, reduced the displayed create-map canvas size, and improved tablet page scrolling.

2026-07-20: Added detector bridge startup helpers, Gradle launch tasks, and automatic emulator detector connection defaults.

2026-07-21: Cleaned up the active map page, added a floating detector reading card, added detector Record capture, and displayed calculated mrem/hr exposure rates. Corrected front-page point scaling, added contamination and airborne table calculations, added control-scheme metadata/filters, and refined generated front-page metadata layout.

In progress:

Final export/share package completion, final physical hardware calibration/validation, showcase dry run, and SME review remain as final handoff tasks.

SIP Brief Part 3

Evaluation Plan

The project evaluation was conducted through a combination of build verification, manual workflow testing, persistence checks, generated-document review, and detector data validation. Each meaningful implementation change was compiled with Gradle using `compileDebugKotlin`, and major changes were recorded in the project change log with scope, files changed, verification, and rollback notes. Application behavior was evaluated in the Android Studio emulator and through tablet-oriented UI review by walking through the primary field workflow: opening the app, creating or importing a map template, starting a new survey from a template, entering survey metadata, placing survey points, recording detector readings, saving the job, reopening the saved record, and confirming that map points, tables, notes, signatures, metadata, and generated front/back page previews remained intact.

Sensor and hardware-related evaluation used a serial-to-TCP detector bridge so CPM readings could be delivered to the Android application while running in the emulator. The live detector card was checked for CPM display, mrem/hr conversion, target-field selection, and Record button behavior. Radiation survey points were verified to receive calculated exposure-rate values while contamination and airborne workflows preserved CPM values for table calculations. Contamination and airborne evaluation focused on background subtraction, <MDA threshold behavior, pCi/uCi/ml derivations, and generated back-page rendering. Offline operation was evaluated by confirming that all core map, survey, record, filter, and review workflows use local Room/SQLite storage without requiring wireless communication or cloud services.

Project Completion Assessment

Status as of 2026-07-22: The project has completed the core MVP and moved through SIP409 hardening focused on template management, record integrity, detector-reading capture, exposure-rate conversion, contamination and airborne calculations, control-scheme metadata, generated front/back page accuracy, diagram-aligned page layouts, and tablet usability polish. The prototype now demonstrates the intended end-to-end workflow: reusable map templates, active survey recording, local persistence, record reopening/editing, generated survey pages, and detector-assisted data entry.

Completed milestones with dates: 2026-05-27 established the Compose UI shell, Room persistence, repository wiring, and baseline map workflow; 2026-05-28 implemented front/back page handling, draft-first state, point constraints, and erase support; 2026-05-30 completed front/back rendering and editable contamination/airborne table workflows; 2026-06-02 validated save/load integrity, draft commits, bulk persistence, and reliable blank-map rendering. 2026-07-12 added the canvas-based map creator, separated map templates from survey records, enforced template-only draft behavior until Save Job, moved notes/signatures to survey scope with Room version 5, and expanded canvas shape tools. 2026-07-16 through 2026-07-22 improved canvas tool visibility, aligned Home/Maps/Records with planned layouts, added record image previews, introduced shared visible scroll rails, and added detector bridge startup tooling. Added the floating live detector card, detector Record capture, centralized CPM-to-mrem/hr conversion, front-page point-scaling correction, contamination/airborne calculation support, control-scheme metadata/filtering, and front-page metadata layout refinement. Remaining work is limited to final export/share package polish, physical detector calibration/validation before operational use, final SME review, and showcase delivery.

The project should be considered a functional academic prototype rather than an operational radiation protection instrument. The software workflow, persistence model, detector-reading capture, and calculation logic have been implemented and repeatedly compiled successfully, but final operational use would require formal detector/probe calibration, procedure review, cybersecurity review, and approval by qualified radiological controls personnel.

Appendices

Appendix A: Innovation brief PDF – [SIP311_Innovation_Brief.pdf](#)

Appendix B: RadconApp change log – [docs/change-log.md](#)

Appendix C: Software architecture documentation – [docs/software-architecture.md](#)

Appendix D: Requirements traceability documentation – [docs/requirements-traceability.md](#)

Appendix E: Showcase speech outline – docs/showcase-speech-outline.md

Appendix F: Android/Kotlin source project – RadconApp app/src/main/java/com/radsurvey/app

Appendix G: Prototype UI screenshots – home_page.jpg, maps_pages.jpg, map.jpg, records_page.jpg

Appendix H: Top-level design image – top_level_design.jpg

Appendix I: Prior art reference – Thermo Fisher Scientific RadEye PRD product page

Appendix J: Build verification evidence – Gradle compileDebugKotlin checks recorded in docs/change-log.md